

Principiile SOLID Laborator

Tehnici avansate de programare

Nicoleta Scrimint



Facultatea de
MATEMATICĂ



C# certification

- <https://www.freecodecamp.org/learn/foundational-c-sharp-with-microsoft/#write-your-first-code-using-c-sharp>



Email course for SOLID

- <https://mailchi.mp/nimblepros/solid-email-course>



SOLID video

- <https://learn.microsoft.com/en-us/shows/visual-studio-toolbox/solid-design-patterns>



Learning sources I

- <https://deviq.com/practices/pain-driven-development>
- <https://learn.microsoft.com/en-us/archive/msdn-magazine/2008/june/patterns-in-practice-the-open-closed-principle>
- <https://www.thoughtworks.com/insights/blog/agile-engineering-practices/solid-principles-how-to-create-a-code-that-is-easy-to-extend-and-maintain-part-1>
- <https://www.pluralsight.com/courses/principles-oo-design>
- <https://www.pluralsight.com/courses/csharp-solid-principles>
- <https://github.com/ardalis/SolidSample>
- <https://deviq.com/principles/solid>
- <https://enterprisecraftsmanship.com/posts/ocp-vs-yagni/>
- <https://enterprisecraftsmanship.com/posts/most-valuable-software-development-principles/>



Learning sources II

- <https://martinfowler.com/bliki/TellDontAsk.html>
- <https://www.thoughtworks.com/insights/blog/agile-engineering-practices/solid-principles-how-to-create-a-code-that-is-easy-to-extend-and-maintain-part-1>
- <https://www.thoughtworks.com/insights/blog/agile-engineering-practices/solid-principles-how-to-create-a-code-that-is-easy-to-extend-and-maintain-part-2>
- <https://www.thoughtworks.com/insights/blog/agile-engineering-practices/solid-principles-how-to-create-a-code-that-is-easy-to-extend-and-maintain-part-3>
- <https://martinfowler.com/articles/dipInTheWild.html>
- <https://martinfowler.com/articles/injection.html>
- <https://enterprisecraftsmanship.com/posts/singleton-vs-dependency-injection/>
- <https://martinfowler.com/articles/dipInTheWild.html>
- <https://learn.microsoft.com/en-us/dotnet/architecture/modern-web-apps-azure/architectural-principles#dependency-inversion>



Exercises

- <https://github.com/ardalis/SolidSample>
- Choose a SOLID principle and fix it
- You can also inspire from the solution under final version



Mandatory homework I

- Create a console application for E-commerce Order
- Functional features
 - Product catalog
 - Shopping cart
 - Orders – can ship with multiple payment systems
 - Discounts – future different discount campaigns will be applied
 - Shipping – can ship with multiple courier companies
- Identify the end users of the software system



Mandatory homework I

- Create the C4 model diagram for the software system
 - System context diagram
 - Container diagram
- <https://c4model.com/>

